
Continuous PHQ-9 Regression from Mandarin Speech: A Task- and Valence-Stratified Analysis on MODMA

Ashwini Athreya Alicia Danielle Rasika Ramanan Arianna Yuan

Abstract

We use the multimodal open dataset for mental disorder analysis (MODMA) to predict PHQ-9 depression severity as a continuous score from acoustic and linguistic features extracted from audio files. After comparing several models, our primary model is Support Vector Regression with Elastic Net feature selection. Features come from two pipelines: 88-dimensional eGeMAPSv02 acoustic functionals via openSMILE and Whisper-derived linguistic features. Using leave-one-subject-out (LOSO) cross-validation across all 52 subjects (23 MDD, 29 HC), the best configuration (combining both feature streams with demographics) achieves an RMSE of 6.81 PHQ-9 points, which is a 1.75-point improvement over the mean-predictor baseline of 8.56. Stratifying by elicitation task shows the depression signal is distributed across MODMA’s elicitation set rather than concentrated in any single task. Stratifying by emotional valence reveals that positive-valence prompts are the only configuration where linguistic features outperform acoustic ones, reflecting the anhedonia signal targeted by these prompts. Post-hoc multicalibration substantially reduced group-conditional prediction bias, shrinking the MDD and HC mean prediction gaps from 5.79 to 0.42 and from 3.56 to 0.51 PHQ-9 points, respectively. Leave-one-out conformal 95% prediction intervals attain 92.3–94.2% empirical marginal coverage (quantized at $n = 52$). To our knowledge, this is the first attempt at continuous PHQ-9 regression on MODMA.

1. Introduction

Major depressive disorder (MDD) affects roughly 280 million people worldwide and is routinely under-diagnosed. Speech and EEG biomarkers are an active area for inexpensive, scalable screening. Almost all published machine learning work on the multimodal open dataset for mental disorder analysis (MODMA) (Cai et al., 2022) frames the problem as binary MDD-versus-healthy classification, with

reported classification accuracies exceeding 90% (Yousufi et al., 2024). A categorical label is, however, an unrefined method: a clinician choosing between watchful waiting, brief intervention, and urgent referral cares about the gradient of symptom severity, not just whether a participant crosses a diagnostic threshold. The PHQ-9 is a 9-item self-report questionnaire (items in Appendix 5) scored 0–27 that maps directly to the DSM diagnostic criteria for major depressive disorder (Kroenke et al., 2001). Each item rates how often, over the past two weeks, the respondent has been bothered by a specific symptom on a 0–3 scale; conventional thresholds are ≥ 5 for mild, ≥ 10 for moderate, and ≥ 20 for severe depression.

Our central research question is: *can acoustic and linguistic features from Mandarin speech predict PHQ-9 severity on a continuous scale, and how does the predictive signal vary across speech elicitation tasks?* We break this into two sub-questions:

- **RQ1.** Can speech features alone predict the continuous PHQ-9 severity score, and can we attach a calibrated prediction interval to each estimate?
- **RQ2.** Which of MODMA’s speech elicitation tasks (clinical interview, word reading, passage reading, picture description) and which emotional valences (positive, neutral, negative) give the strongest signal?

1.1. Related Work

Speech and EEG biomarkers of depression. Frontal alpha asymmetry is an established neural correlate of depression (Van der Vinne et al., 2025), and acoustic biomarkers (prosody, spectral flatness, voice quality) are validated predictors of depression and suicide risk (Cummins et al., 2015). Continuous-severity regression from speech is the standard task on the English DAIC-WOZ corpus (Gratch et al., 2014), where multimodal fusion of acoustic and linguistic features improves PHQ-8 RMSE by 1–2 points.

Prior MODMA work. None of the 48 published MODMA papers we surveyed frame the problem as PHQ-9 regression; all treat it as binary classification. The closest precedent is Tasci et al. (Tasci et al., 2023), who report 84% binary classification accuracy under leave-one-subject-out evaluation,

which is realistic compared to the inflated within-subject splits common in the literature.

Our contribution. We treat this gap as the problem: a calibrated, honest report of how well one can do continuous PHQ-9 prediction at $n = 52$, a defensible evaluation protocol, and a fine-grained look at which speech elicitations carry the most signal.

2. Methods

2.1. Dataset and Labels

Participant Recruitment. The MODMA dataset (Cai et al., 2022) was collected at Lanzhou University Second Hospital, Gansu, China. Patients with major depressive disorder (MDD) were recruited from the hospital’s inpatient and outpatient populations, each diagnosed and recommended by at least one clinical psychiatrist; healthy controls (HC) were recruited from the local community using posters. Figure 4 shows the analyzed sample’s age, gender, and education distributions: the sample is predominantly male, and most subjects have at least 16 years of education.

Inclusion Criteria. All participants had to be 18–55 years old with at least primary education, no past-year alcohol or psychotropic drug abuse or dependence, and (for women) not pregnant, lactating, or on birth control. MDD inclusion additionally required a Mini-International Neuropsychiatric Interview meeting DSM-IV depression criteria, a Patient Health Questionnaire-9 (PHQ-9) score ≥ 5 (Figure 1, dotted line), and no psychotropic medication in the prior two weeks. HC inclusion required no personal or family history of mental disorders. The resulting cohort contains 23 MDD and 29 HC subjects.

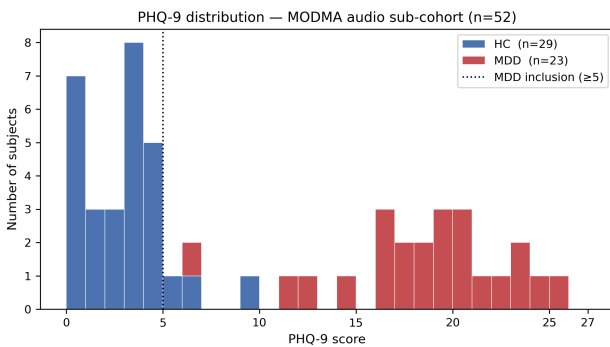


Figure 1. PHQ-9 distribution in the audio sub-cohort ($n = 52$). The dotted line marks $\text{PHQ-9} \geq 5$, one of the inclusion criteria for MDD.

Experiments Conducted. EEG was collected under two protocols: a 128-channel HydroCel cap during eyes-closed resting state and an emotional dot-probe task, and a 3-electrode wearable device during resting state only. Audio

was collected across four spoken tasks: interview, passage reading, word reading, and picture description.

Experiments Used for Analysis. This project uses only the audio modality, available for $n = 52$ subjects (23 MDD, 29 HC) — our full sample. Each subject contributed 29 audio recordings across four tasks; Table 1 gives the task breakdown, per-task counts, valence labelling, and task descriptions.

2.2. Feature Extraction

We extract two complementary feature types from each audio recording — acoustic and linguistic — to capture two distinct channels through which depression manifests in speech (Cummins et al., 2015; Eichstaedt et al., 2018). Their extraction pipelines are described below and illustrated in Figure 5.

Acoustic features (openSMILE). We extract acoustic features using the openSMILE toolkit (Eyben et al., 2010) configured with the eGeMAPSv02 feature set (Eyben et al., 2016). eGeMAPSv02 is a standardized 88-functional acoustic parameter set developed for voice research and affective computing; it specifies both the low-level acoustic descriptors to extract and the functionals to compute over them. Following this specification, openSMILE samples the descriptors continuously over each recording and aggregates them into 88 recording-level functionals.

Linguistic features (Whisper). We transcribe each recording to Mandarin Chinese text using the Whisper-large-v3 model (Radford et al., 2023), run via the faster-whisper implementation. Each transcript is then cleaned via NFKC Unicode normalization, traditional-to-simplified Chinese conversion, punctuation restoration, and word- and sentence-level segmentation — producing a tokenized form ready for feature extraction. From each cleaned transcript we extract 16 features grouped into three families: 5 *lexical* features (rates of first-person singular and plural pronouns, negation tokens, the type-token ratio, and MATTR-50); 3 *syntactic* features (mean and standard deviation of word-tokens per sentence, and punctuation density); and 8 *sentiment* features (rates of the seven DUTIR macro emotion categories — joy, like, anger, sadness, fear, disgust, and surprise — plus a net-polarity score, all computed from the DUTIR Chinese Emotion Vocabulary Ontology (Xu et al., 2008)).

For each subject and analysis subset, we aggregate per-recording features into a *single subject-level vector* by taking the per-feature mean across the subject’s available recordings in that subset. Missing or excluded recordings (corrupt acoustic files for one subject, unusable transcripts for another) are dropped from the affected analyses.

Table 1. Audio recording structure by task.

Task	Description	Recordings / subj.	Valence			
			POS	NEG	NEU	NA
Interview	Free response to one interview question.	18	6	6	6	0
Passage Reading	Read aloud “The North Wind and the Sun”: a fable (Chinese version) used worldwide in speech research.	1	0	0	0	1
Word Reading	Read aloud of 10 common Chinese words selected from Hongfei Lin’s affective ontology corpus and the Chinese sentimental words extremum table.	6	2	2	2	0
Picture Description	Free response descriptions of three face image (each valenced), plus a “crying woman” scene from the Thematic Apperception Test.	4	1	1	1	1

Note. Valence shorthand: POS = Positive, NEG = Negative, NEU = Neutral, NA = Unlabelled.

2.3. Nested LOSO with SVR

Our primary model is Support Vector Regression (SVR) with an RBF kernel. Fig. 2 summarizes the full procedure: outer LOSO (k) wraps a 32-combination grid search (g) over SVR hyperparameters $C \in \{0.1, 1, 10, 100\}$, $\epsilon \in \{0.1, 0.5, 1.0, 2.0\}$, and $\gamma \in \{1/(d \cdot \text{Var}(X)), 1/d\}$ (Eqs. 2 and 3), which in turn wraps an inner LOSO over the 51 outer-training subjects (j); the pipeline is fit per inner fold. Within each pipeline fit, missing values in the inner-training matrix are first imputed with each feature’s training-set median, and every feature is then standardized to zero mean and unit variance using the 50 inner-training subjects’ data. The result then passes through an Elastic Net (Eq. 1) acting as a feature filter: it fits regression coefficients and drops any feature whose coefficient is zero. Its hyperparameters $\alpha = 0.1$ and ℓ_1 -ratio 0.7 are held fixed rather than tuned inside the nested CV. SVR fits an RBF-kernel regression on the surviving features and predicts the inner-held-out subject; once outer fold k ’s 32×51 inner predictions complete, the combination g^* that minimizes mean inner-LOSO RMSE is refit on all 51 outer-training subjects via the full pipeline and used to predict the outer-held-out subject k . A conformal 95% prediction interval is attached to each outer-fold prediction as described in §2.4.

2.4. Prediction intervals

For RQ1 we attach 95% prediction intervals to each held-out LOSO prediction via leave-one-out conformal calibration. The absolute LOSO residuals form the calibration set, and we take the finite-sample-corrected $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of these residuals as the interval radius (Vovk et al., 2005), yielding a constant-width interval with a formal coverage guarantee under exchangeability. We apply the procedure to the all-task model and to each valence and task subset.

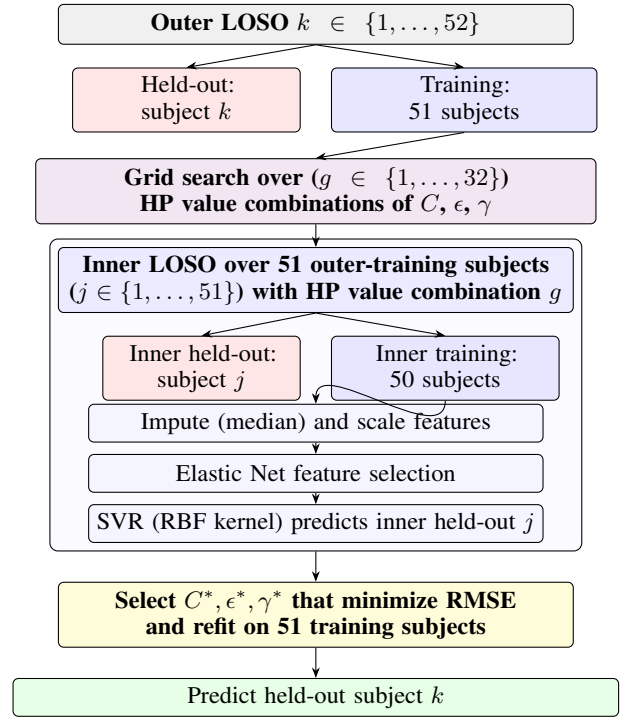


Figure 2. Nested leave-one-subject-out evaluation pipeline. HP abbreviates hyperparameter.

2.5. Multicalibration

We applied a multicalibration-inspired post-hoc subgroup correction procedure, adapted from the iterative multicalibration framework of Hébert-Johnson et al. (Hébert-Johnson et al., 2018), to the final EN+SVR PHQ-9 predictions. Unlike the original multicalibration formulation, which was developed for probabilistic classification, our adaptation operates directly on continuous PHQ-9 regression scores. This step does not modify the eGeMAPS, Whisper, or demographic features; instead, it operates only on

the OOF predicted PHQ-9 scores produced by the final `egemaps_whisper_demo` model. Calibration groups were defined using diagnostic group (MDD vs. HC), gender, prediction-score quantile bins, and intersections between metadata groups and score bins. For each subgroup, we computed the mean residual between the true and predicted PHQ-9 scores. At each iteration, the subgroup with the largest absolute mean residual was selected, and predictions for members of that subgroup were shifted toward the observed subgroup mean. Iterative updates continued until the largest remaining subgroup residual fell below a fixed tolerance or until a maximum number of iterations was reached. Because calibration was performed exclusively on OOF predictions, the procedure avoided direct access to training-fold targets during subgroup correction. We evaluated the effect of post-hoc calibration by comparing RMSE, MAE, R^2 , and subgroup mean prediction gaps before and after calibration.

2.6. Baselines and metrics

Our floor is the mean-predictor baseline: predict $\bar{y}_{\text{train}}$ for every held-out subject (where the mean is taken over the 51 training subjects in each LOSO fold). Any model with test $\text{RMSE} \geq \text{baseline}$ is adding no value. We report RMSE, MAE, and R^2 aggregated across the 52 LOSO held-out predictions. Coverage and mean interval width are reported for the prediction intervals. Per-group calibration error is reported for the multicalibration analysis.

3. Results and Discussion

3.1. Feature-stream ablation

Table 2 compares SVR test performance across six feature configurations against the mean-predictor baseline (which all six beat). Acoustic features outperform linguistic ones: eGeMAPS alone reaches RMSE 7.46 vs. Whisper’s 7.98, and combining the two streams (RMSE 7.37) yields only a 0.09-point lift over the better single stream, indicating marginal complementarity. Adding demographic covariates (age, sex, years of education) drives the largest gains, with the best configuration — eGeMAPS + Whisper + demographics — reaching RMSE 6.81 ($R^2 = 0.34$).

Table 2. SVR feature-stream ablation across LOSO predictions ($n = 52$). Best test RMSE in bold.

Configuration	Features	RMSE	MAE	R^2
Mean baseline	0	8.56	7.97	-0.04
eGeMAPS	88	7.46	6.03	0.21
+ demographics	91	6.89	5.59	0.33
Whisper	16	7.98	6.72	0.10
+ demographics	19	7.64	6.49	0.17
eGeMAPS + Whisper	104	7.37	6.13	0.23
+ demographics	107	6.81	5.66	0.34

3.2. Effect of Multicalibration on EN+SVR

Table 3 compares the final `egemaps_whisper_demo` EN+SVR predictions before and after multicalibration across the 52 LOSO held-out predictions. Post-hoc multicalibration substantially reduced subgroup-level residual bias while also improving overall prediction accuracy. The MDD subgroup’s absolute mean residual decreased from 5.79 to 0.42 PHQ-9 points, while the HC subgroup’s residual decreased from 3.56 to 0.51 points. Rather than sacrificing overall accuracy to reduce subgroup bias, post-hoc multicalibration improved both fairness and predictive performance. The reduction in subgroup residuals was accompanied by a decrease in overall RMSE from 6.81 to 3.99 PHQ-9 points, suggesting that the raw EN+SVR predictions contained systematic subgroup-specific error that could be corrected without retraining the model.

Table 3. Effect of multicalibration on final `egemaps_whisper_demo` EN+SVR predictions. Per-group absolute mean residual is reported separately for MDD and HC subgroups.

	RMSE	MDD bias	HC bias
EN+SVR (raw)	6.81	5.79	3.56
EN+SVR (multicalibrated)	3.99	0.42	0.51

3.3. Performance by emotional valence

27 of the 29 recordings carry a valence label assigned during stimulus design: 9 positive, 9 neutral, 9 negative. The two unlabeled recordings (passage reading and the TAT scene) are excluded.

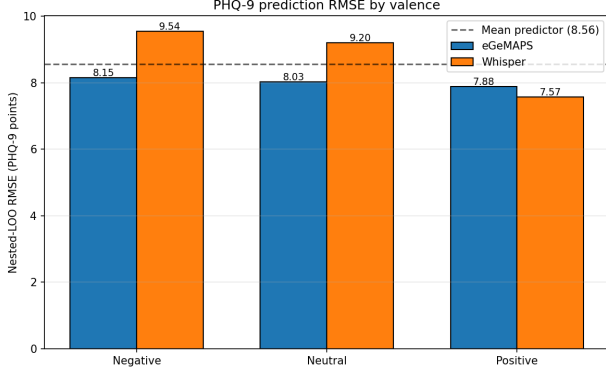


Figure 3. Nested-LOO RMSE by emotional valence and feature modality. The eGeMAPS-vs-Whisper ranking inverts on positive valence: positive is the only configuration where Whisper transcript features outperform eGeMAPS acoustics on the same data.

Positive valence is the only configuration in the ablation where the modality ranking reverses (Figure 3), full metrics in Table 7. Whisper transcript features outperform eGeMAPS acoustics on the same 9-file subset (RMSE = 7.57 vs 7.88), and Whisper-positive outperforms the mean-predictor baseline by a full PHQ-9 point. The pattern does not transfer to the other two valences, where Whisper falls well below baseline (RMSEs 9.20 and 9.54, $R^2 < 0$) while eGeMAPS retains small positive R^2 values across all three.

This pattern is consistent with the clinical structure of the PHQ-9. Question 1 measures anhedonia: “little interest or pleasure in doing things”, and positive-valence prompts (e.g., “describe your happiest moment”) are designed to elicit affectively engaged responses that target precisely this symptom. Prior work has documented that depressed speakers produce shorter, more negation-heavy, and less affectively elaborated responses to positive emotional content (Cummins et al., 2015; Stasak et al., 2017), which is precisely what the 16-dimensional Whisper feature set captures (first-person rate, negation rate, type-token ratio, sentence-final particles, net polarity). Negative-valence prompts elicit rich responses from both cohorts, so transcript-level features lose their discriminative power.

The eGeMAPS positive-valence run also achieves a 95% conformal PI width of 25.20 PHQ-9 points, the tightest non-demographic interval in the entire ablation.

3.4. Performance by elicitation task

The MODMA audio data spans four elicitation types: an 18-question clinical interview, 6 word readings, a single passage reading (“The North Wind and the Sun”), and 4 picture descriptions (3 pictures plus 1 TAT scene). To test whether any single elicitation carries the depression signal disproportionately, we restrict feature aggregation to each

task’s files and rerun the full nested-LOO SVR pipeline on each subset, keeping the model class, hyperparameter grid, and feature axes fixed.

Table 4. Per-task SVR with eGeMAPS or Whisper features. File counts: Interview 18, Word Reading 6, Passage Reading 1, Picture Description 4. Mean-predictor baseline RMSE = 8.56; bold beats baseline. Every eGeMAPS subset clears baseline; no Whisper subset does.

Task	Modality	RMSE	MAE	R^2	PI width
Interview	eGeMAPS	7.95	6.91	0.10	26.85
	Whisper	8.76	7.51	-0.09	32.69
Word Reading	eGeMAPS	7.97	6.92	0.10	28.07
	Whisper	9.44	7.90	-0.26	33.22
Passage Reading	eGeMAPS	8.18	7.09	0.05	27.71
	Whisper	9.22	7.42	-0.20	33.99
Picture Desc.	eGeMAPS	8.24	7.27	0.04	26.58
	Whisper	9.06	7.62	-0.16	34.42
All tasks (ref.)	eGeMAPS	7.46	6.03	0.21	26.94

The eGeMAPS subsets cluster tightly: the gap between best (Interview, 7.95) and worst (Picture Description, 8.24) is 0.29 PHQ-9 points, which is within the sampling variation at $n = 52$. None of the four single-task eGeMAPS models match the all-tasks model at 7.46, indicating that the depression signal in MODMA’s audio is distributed across the elicitation set rather than concentrated in any one task type. This is only partially consistent with prior elicitation work (Stasak et al., 2017), which has shown that the relative discriminability of elicitation types varies with the feature set and corpus. The practical implication is that shortening the screening protocol by dropping elicitations would lose signal that cannot be recovered.

Every single-task Whisper run performs worse than the mean-predictor baseline, with R^2 values ranging from -0.09 (Interview) to -0.26 (Word Reading). With only 16 transcript-derived features and 1–6 files per subject in a slice, the per-subject mean transcript carries too little information for the SVR to learn from at $n = 52$. The 33–34-point prediction interval widths (wider than the mean-predictor baseline’s 27.65) reflects this.

3.5. Prediction interval calibration and width

We attach a 95% conformal prediction interval to every out-of-fold prediction using the leave-one-out construction. We investigate two questions: are the intervals actually calibrated, and which feature combinations make them tighter?

Calibration. At $n = 52$, empirical coverage can only take values in increments of $1/52 \approx 1.9\%$, so an exact 95% rate is unreachable. The closest achievable values are $48/52 = 92.3\%$ and $49/52 = 94.2\%$. Every one of our 21 runs lands

on one or the other (12 at 94.2%, 9 at 92.3%), consistent with the finite-sample coverage guarantee for jackknife-conformal intervals under exchangeability (Barber et al., 2021).

However, marginal coverage masks a clinically important group-level asymmetry. Splitting coverage by group, MDD subjects are systematically under-covered across the 21 runs (mean 90.9%) against HC coverage of 95.4%, which is a 4.5-point gap (Table 5). The gap is largest where the model is weakest. The mean predictor, the Whisper-only baseline, and the failing Whisper task subsets all sit at 82.6% MDD coverage against 100% HC coverage. These are runs where the model is barely doing more than predicting the cohort mean, so MDD predictions are systematically far from the actual MDD scores. The marginal calibration set is dominated by these large MDD residuals, but the 95th-percentile radius remains insufficient to reach the most severe MDD subjects. On the strongest runs (egemaps_whisper_demo, egemaps_demo) the direction reverses: HC becomes slightly under-covered at 93.1%, because the model has learned to fit MDD well enough that the MDD residuals shrink below the HC distribution.

This failure mode is what motivates the multicalibration framework (Hébert-Johnson et al., 2018): marginal calibration on a bimodal cohort can mislead about reliability for the clinically high-stakes group. Mondrian (group-conditional) conformal (Vovk et al., 2003) would address this by computing the prediction-interval radius from each group’s residuals separately, so MDD subjects’ intervals reflect MDD residual variability and HC subjects’ intervals reflect HC variability. Each group would then meet the 95% coverage target within each group. The group-conditional diagnostic also doubles as an indirect model-quality check: runs that achieve marginal coverage by predicting near the cohort mean fail group-conditionally, while runs that genuinely learn the depression signal converge toward balanced coverage across groups.

Width. Width varies substantially across configurations, ranging from 25.09 (whisper_demo) to 34.42 (whisper_picture_description_only), which is a 9.3-point spread across runs that all maintain valid marginal coverage (Table 5).

Demographics tighten the interval more reliably than any other intervention. Adding age, gender, and education years to any feature stack contracts the PI: whisper_only’s 28.19 drops to 25.09 in whisper_demo, a 3.1-point move that produces the tightest interval in the table. eGeMAPS contracts less because it starts tighter (26.94 to 25.71), and the combined acoustic-linguistic stack contracts by another point when demographics are added (27.62 to 26.61). Education years carries the

top permutation importance in every demo-containing run, suggesting that demographics explain inter-subject variance the audio features cannot capture on their own.

The lowest-RMSE run is not the tightest-PI run. egemaps_whisper_demo wins on RMSE at 6.81 but is fourth on width (26.61); whisper_demo is tightest at 25.09 with a worse RMSE of 7.64. RMSE is dominated by average squared error, while jackknife-conformal width is determined by the 95th percentile of $|y - \hat{y}|$ and dominated by the worst-case tail. A model can reduce its mean error while widening that tail outward.

The strongest non-demographic interval is egemaps_positive_only at 25.20, second only to whisper_demo by 0.11 points. Read alongside §3.3, this identifies positive-valence speech as the most informative piece of the elicitation set: it yields both the only Whisper subset that beats baseline and an acoustic interval competitive with the demographic-augmented runs.

The Whisper task subsets cluster at the wide end of the table from 32.7 to 34.4 points, which is wider than even the trivial mean-predictor interval at 27.65. An interval wider than the trivial baseline’s indicates that the calibration procedure has correctly identified that the model learned no useful signal. Their RMSEs are also worse than baseline.

4. Process

This section reports the model-comparison evidence on variants we tried before proceeding with the SVR-with-Elastic-Net-feature-selection pipeline used in Section 3.

Baseline. We established a mean-predictor baseline—predicting the training-set mean PHQ-9 for every test subject—yielding a test RMSE of 7.846. This served as the floor any learned model had to clear.

Default and tuned SVR (all 88 features). A default RBF-SVR underperformed the baseline (test RMSE 8.360, $R^2 = -0.14$). Hyperparameter tuning via LOSO grid search worsened test generalization further (RMSE 8.677, $R^2 = -0.23$) despite achieving train $R^2 = 0.692$ —a clear sign of overfitting at $n = 41$ training subjects.

Dimensionality reduction (PCA + SVR). Reducing the 88 eGeMAPS features to 5, 10, or 20 PCA components before RBF-SVR did not recover generalization; all three variants (test RMSE 8.15–8.33) remained above the mean-predictor floor.

Linear SVR. LinearSVR was the worst-performing variant overall (test RMSE 10.07, $R^2 = -0.66$), confirming that a linear boundary in the original feature space is unsuitable for this target.

Tree ensembles. Random Forest and Gradient Boosting

Table 5. Conformal 95% prediction intervals across selected SVR configurations, sorted by PI width. Marginal coverage is reported alongside coverage on the MDD ($n = 23$) and HC ($n = 29$) subgroups; the marginal rate masks systematic MDD under-coverage on the weaker models. Width in PHQ-9 points.

Configuration	RMSE	Marg.	MDD	HC	PI width
whisper_demo	7.64	94.2%	91.3%	96.6%	25.09
egemaps_positive_only	7.88	94.2%	91.3%	96.6%	25.20
egemaps_demo	6.89	94.2%	95.7%	93.1%	25.71
egemaps_whisper_demo	6.81	94.2%	95.7%	93.1%	26.61
egemaps_only	7.46	94.2%	95.7%	93.1%	26.94
mean_predictor	8.56	92.3%	82.6%	100%	27.65
whisper_only	7.98	94.2%	87.0%	100%	28.19
whisper_pict_desc_only	9.06	92.3%	82.6%	100%	34.42

(both LOSO-tuned) similarly failed to clear the baseline: RMSE 8.38 and 9.01, respectively.

Elastic Net as a direct predictor. ElasticNetCV with LOSO inner cross-validation over eight ℓ_1 -ratio values and 40 log-spaced λ values produced a test RMSE of 9.35—worse than all SVR variants despite strong regularization.

Pivot: Elastic Net as feature selector. Although Elastic Net was a weak direct predictor, its sparse solution retained 77 of 88 features. Re-training RBF-SVR on only those 77 features (LOSO-tuned) was the only configuration to beat the baseline (Table 6). This two-stage EN $\rightarrow$ SVR pipeline was adopted as the final model for the subset analyses and prediction-interval experiments reported in Section 3.

Takeaway. Heavy tuning of a single-stage model reliably increased train–test gap under $n = 41$. The lesson is that sparse linear pre-selection stabilizes a subsequent nonlinear regressor by removing redundant correlated features before the kernel sees them—consistent with the motivation for Elastic Net in correlated feature sets (Zou & Hastie, 2005). The improvement over baseline is small and no Spearman p -value reaches conventional significance ($p_{\min} \approx 0.145$), which we attribute honestly to the 11-subject test set rather than model failure.

Table 6. Model comparison on the held-out test set ($n = 11$). The mean-predictor baseline RMSE is 7.846.

Model	RMSE	R^2	Spearman ρ
Mean baseline	7.846	—	—
SVR, all 88 features (default)	8.360	−0.14	—
SVR, all 88 features (tuned)	8.677	−0.23	0.313
PCA (10 components) + SVR	8.151	−0.09	—
Linear SVR	10.066	−0.66	—
Random Forest, all 88 features	8.377	−0.15	0.304
Gradient Boosting	9.012	−0.33	0.313
Elastic Net (direct)	9.351	—	0.470
RF, 77 EN-selected features	8.046	−0.06	0.452
SVR, 77 EN-selected features	7.740	0.021	0.456

5. Contributions

Ashwini Athreya led the multicalibration analysis, including the formal calibration framework, the per-subgroup residual diagnostics, and the ablation comparing SVR predictions before and after multicalibration.

Alicia Danielle led model selection: the five-model comparison (linear, Ridge, Elastic Net, Random Forest, SVR) on the eGeMAPS feature set under LOSO evaluation, and the rationale for choosing SVR as the primary model with Elastic Net as a supplementary baseline.

Rasika Ramanan led dataset selection and the literature review covering the 48 published papers on MODMA. She also spearheaded feature extraction, implementing the openSMILE eGeMAPSv02 extraction script and the Whisper transcript and linguistic-feature pipeline. Finally, she led the per-participant feature aggregation, the LOSO-with-standardization training harness, and the SVR feature-ablation experiments (eGeMAPS only, Whisper only, combinations, with demographics).

Arianna Yuan led the held-out experiments and uncertainty quantification: the leave-one-out conformal prediction intervals, the per-valence and per-task SVR experiments, and the analysis of which ablations narrow the prediction interval.

Code and Data

Project code, the analysis notebook, and the feature extraction scripts are available on [GitHub](#). The [MODMA dataset](#) is restricted-access from Lanzhou University. Our extracted feature CSVs, which are derived statistics rather than raw audio, are included in the repo for reproducibility of the modeling results.

AI Usage

We used Claude (Anthropic) as an assistant for literature review (summarizing and indexing the 48 published MODMA papers we surveyed) and figure creation. Every numeric

value in the tables comes from a notebook cell that we ran ourselves. Citations were manually verified against original sources.

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Appendix

A. PHQ-9 items

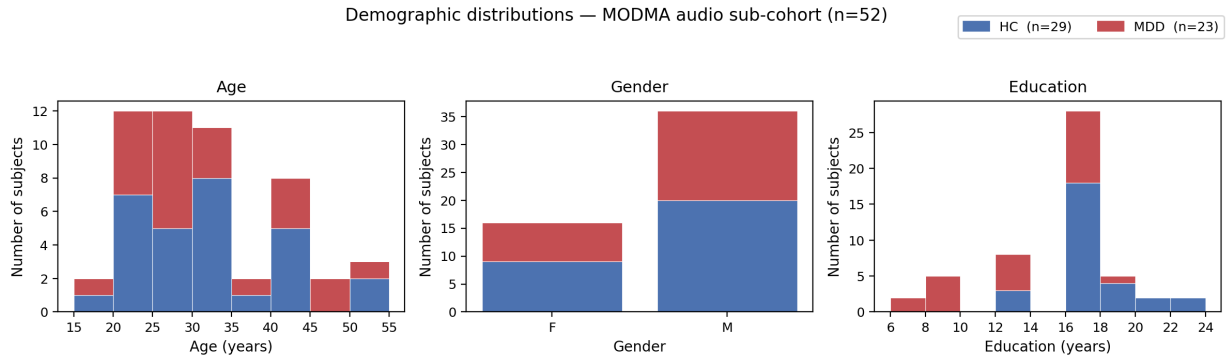
The PHQ-9 (Kroenke et al., 2001) asks the respondent how often over the past two weeks they have been bothered by each of the following 9 symptoms, on a 4-point scale: 0 = *not at all*, 1 = *several days*, 2 = *more than half the days*, 3 = *nearly every day*.

1. Little interest or pleasure in doing things (anhedonia)
2. Feeling down, depressed, or hopeless
3. Trouble falling or staying asleep, or sleeping too much
4. Feeling tired or having little energy
5. Poor appetite or overeating
6. Feeling bad about yourself, or that you are a failure
7. Trouble concentrating on things
8. Moving or speaking slowly, or being fidgety/restless (psychomotor)
9. Thoughts of being better off dead or of self-harm

The 9-item total ranges 0–27. Conventional severity bands are ≥ 5 (mild), ≥ 10 (moderate), ≥ 15 (moderately severe), and ≥ 20 (severe).

B. Demographic distribution of the audio sub-cohort

Figure 4. Age, gender, and education distributions of the audio sub-cohort ($n = 52$), stacked by group.



C. Feature extraction pipeline

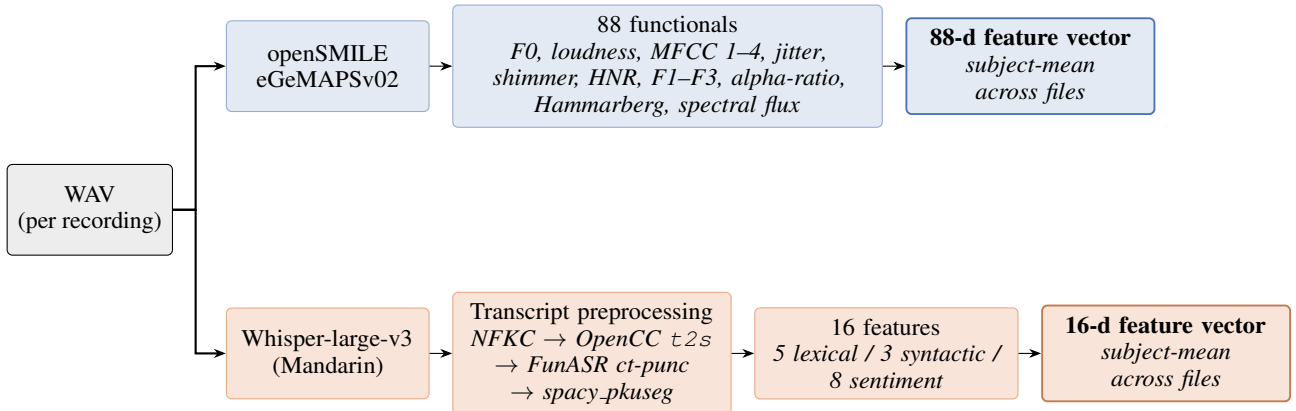


Figure 5. Audio feature-extraction pipeline.

D. Hyperparameter roles in the SVR and Elastic Net objectives

The five hyperparameters tuned and discussed in the nested LOSO pipeline (§2.3) appear in the formulas below in **bold purple**.

Elastic Net feature selection. The Elastic Net step fits coefficients $w \in \mathbb{R}^d$ to training matrix $X \in \mathbb{R}^{n \times d}$ by minimizing:

$$\min_w \frac{1}{2n} \|y - Xw\|_2^2 + \alpha \rho \|w\|_1 + \frac{\alpha(1-\rho)}{2} \|w\|_2^2. \quad (1)$$

Impact of changing each hyperparameter (both fixed in our project): Increasing α increases regularization. Increasing ρ shifts the penalty toward L1 (sparsity) and away from L2 (smooth shrinkage). At our fixed $(\alpha, \rho) = (0.1, 0.7)$, regularization is moderate and sparsity-leaning; features with nonzero Elastic Net coefficients pass downstream to the SVR.

SVR with RBF kernel. Given a training set $\{(x_i, y_i)\}_{i=1}^n$, SVR with ϵ -insensitive loss solves:

$$\begin{aligned} \min_{w, b, \xi, \xi^*} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \\ \text{s.t.} \quad & y_i - \langle w, \phi(x_i) \rangle - b \leq \epsilon + \xi_i, \\ & \langle w, \phi(x_i) \rangle + b - y_i \leq \epsilon + \xi_i^*, \quad \xi_i, \xi_i^* \geq 0 \end{aligned} \quad (2)$$

where $\phi : \mathcal{X} \rightarrow \mathcal{H}$ is the implicit feature map of the RBF kernel and $w \in \mathcal{H}$; the kernel itself is:

$$K(x_a, x_b) = \exp(-\gamma \|x_a - x_b\|^2). \quad (3)$$

Impact of changing each hyperparameter via grid search: Increasing C decreases regularization. Increasing ϵ widens the no-penalty residual band, producing a smoother fit. Increasing γ narrows the RBF kernel, raising variance.

E. Multicalibration post-processing

Multicalibration is applied after the final EN+SVR model has already produced an out-of-fold PHQ-9 prediction $\hat{y}_i$ for each subject. It estimates subgroup-level residual bias and shifts predictions so that each subgroup’s mean prediction better matches its mean observed PHQ-9.

For a subgroup S (e.g., MDD, HC, gender, prediction-score bin, or their intersections), the subgroup residual is

$$r_S = \frac{1}{|S|} \sum_{i \in S} (y_i - \hat{y}_i).$$

At each iteration, the algorithm selects the subgroup with the largest $|r_S|$ and updates predictions for all members of that subgroup:

$$\hat{y}_i \leftarrow \hat{y}_i + r_S, \quad i \in S.$$

Updates stop when the largest subgroup residual is below the tolerance threshold or the maximum number of iterations is reached.

F. Per-valence SVR detailed metrics

Table 7. Per-valence SVR with eGeMAPS or Whisper features (full metrics for Figure 3). Each subset uses 9 files. Bold beats the mean-predictor RMSE of 8.56. Positive valence is the only configuration where Whisper outperforms eGeMAPS on the same data.

Valence	Modality	RMSE	MAE	R^2	PI width
Positive	eGeMAPS	7.88	6.98	0.12	25.20
	Whisper	7.57	5.95	0.19	28.86
Neutral	eGeMAPS	8.03	6.66	0.09	27.77
	Whisper	9.20	7.91	-0.20	29.61
Negative	eGeMAPS	8.15	7.08	0.06	27.60
	Whisper	9.54	8.01	-0.29	32.82